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Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

I have received your dataset: `sicor_municipal_contracts.csv`. I'll first inspect its structure to understand its contents, including column names, data types, and general characteristics. Then, you can ask specific analytical questions based on the dataset.

Let me load and preview the data now.

Dataset Overview:

- 231,636 rows and 19 columns.
- Contains municipal contract data, including funding, investment, commercialization, and industrialization values.
- Key columns:
 - **StateCode, MunicipalityCode** – Identifiers for state and municipality.
 - **MonthIssued** – The month in which the contract was issued.
 - **FundingValue, InvestmentValue, CommercializationValue, IndustrializationValue** – Financial aspects of contracts.
 - **ContractTotalValue** – The overall contract value.

Can the volume of granted contracts be predicted based on the monthly granting history? Define three options to address this issue and select the best one.

Predicting the Volume of Granted Contracts Based on Monthly History

To predict the volume of granted contracts, we need to analyze the dataset's **temporal patterns**. This is a time-series forecasting problem. Below are three potential approaches:

Option 1: Time Series Analysis with ARIMA

- **Method:**
 - Aggregate contracts per month.
 - Use **AutoRegressive Integrated Moving Average (ARIMA)** to model trends and seasonality.
 - Evaluate the model's effectiveness using **AIC/BIC scores** and forecast accuracy.
- **Pros:**
 - Works well with historical trend data.
 - Suitable if the time series is stationary after transformation.
- **Cons:**
 - Struggles with non-linear patterns and external influences.

Option 2: Machine Learning with Regression Models (XGBoost, Random Forest, etc.)

- **Method:**
 - Use features like previous months' contract volume, economic activity, and funding values to train a regression model.
 - Consider models like XGBoost, Random Forest, or Linear Regression.
 - Evaluate using Mean Absolute Error (MAE) or Root Mean Squared Error (RMSE).
- **Pros:**
 - Handles non-linearity better than ARIMA.
 - Can incorporate additional predictors beyond just historical contract volume.
- **Cons:**
 - Needs feature engineering and hyperparameter tuning.
 - May require more data to avoid overfitting.

Option 3: Deep Learning with LSTM (Long Short-Term Memory Networks)

- **Method:**
 - Convert the contract volume data into a sequence-based format.
 - Train an LSTM neural network to capture long-term dependencies.
 - Evaluate using RMSE and loss functions.
- **Pros:**
 - Captures long-term dependencies in time-series data.
 - Handles complex patterns better than ARIMA.
- **Cons:**
 - Computationally expensive.
 - Requires large datasets to generalize well.

Best Approach: Option 2 (Machine Learning Regression)

Reasoning:

- More **flexible** than ARIMA (handles non-linear patterns).
- Easier to **train and interpret** than LSTMs.
- Can include additional variables like **economic activity, investment values, and funding sources**, improving predictive power.

Next Steps:

Would you like me to proceed with implementing Option 2 (Machine Learning Regression)? 🚀

Message Data Analyst



ChatGPT can make mistakes. Check important info.